

CSE 4408

System Analysis and Design

Lab 11: Designing Effective Output

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MedRadar

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Project Context

MedRadar is a location-based medicine locator system aimed at helping students and local residents quickly find pharmacies that have the required medicines in stock.

Key System Output

Chosen Output	User	Purpose	Delivery Method
Medicine Search Results Page	Customers	To allow customes to quickly medicines in nearest pharmacies	Screen
Order Confirmation & Status Updates	Customers	To provide assurance that the order was received	Screen/ Print
Pharmacy Stock Dashboard	Pharmacy	To manage inventory in real time, stock query etc	Screen
Low Stock Alert / Exception Report	Pharmacy	To highlight medicines running low or nearing expiry	Notification
Sales & Demand Trend Visualization	Pharmacy	To track demand patterns, analyze sales	Screen/ PDF Export

Key System Output

Objective	MedRadar Justification
Serves Purpose	Helps customers find medicines and pharmacies; helps pharmacies track stock and demand.
User-Fit	Customer screens show searches and orders; pharmacy dashboards show inventory, alerts, and trends.
Right Quantity	Displays only relevant medicines, orders, stock alerts, or sales trends.
Where Needed	Accessible via web/mobile screens, dashboards, notifications, and reports.
On Time	Real-time updates for stock, alerts, and order status; weekly or instant trend updates.
Right Method	Screens for searches/orders, dashboards for stock/trends, alerts for low stock, PDF reports for summaries.

Key System Output

Selected Output for Mockups

- Medicine Search Results Page
- Pharmacy Dashboard
- Customer Order Status Page

UI Mockup 1

Medicine Search Result

Dashboard

Order Status

View Cart

Location

Pharmacy Near Me

Select Division

Select District

Select City

Medicine Category

☒ Pain Relief

☐ Antibiotics

☐ Ointment

☐ Vitamins

☐ Other

Price Range

Min

Max

Clear All Filters

[Home](#) [Profile](#) [About Us](#) [FAQ](#)

MedRadar
The Radar to your Remedy



Cart (0)

A

search result with filter 'Pain Relief'

Sort By: Price (Low to High)

7+ results found



A tk

Antacid Tablets

In Stock

1.1 miles away @

View Details

Add to Cart



A tk

Napa Syrup

In Stock

11.3miles away @

View Details

Add to Cart



A tk

Tufnil Tablets

In Stock

2 miles away @

View Details

Add to Cart



A tk



A tk



A tk

Core Output 1

Medicine Search Result

Data Elements:

- Medicine_Name, Medicine_ID, Category_Code
- Pharmacy_ID, Location_Info
- Quantity (Stock_Record)
- Order_Status
- Customer_ID (backend)

Design Choices:

- Search bar + filters (Category, Location, Price)
- Card layout with image, name, price
- Stock status + distance highlighted
- Action buttons (“View Details”, “Add to Cart”)
- Sorting options
- Easy navigation at header

UI Mockup 2

Pharmacy Dashboard

- Dashboard
- Inventory Management
- Order Processing
- Alerts & Notifications
- Reports & Analytics

Pharmacy Dashboard

Welcome, XYZ

Pending Orders 7

Total Medicine Stock

1,245

All products in inventory →

Low Stock Items

12

Items below reorder point →

Expiring Soon

8

Next 30 days →

Out of Stock

2

Urgent reorder needed →

Detailed Medicine Inventory

Filter by Category ▼ Filter by Stock Level ▼ Filter by Expiry Date Clear Filters

Medicine Name	Batch	Expiry Date	Quantity
Paracetamol 500mg	P-12345	2025-10-20	250
Amoxicillin 250mg	A-67890	2024-08-15	50
Ibuprofen 200mg	I-09876	2026-03-01	300

Core Output 2

Pharmacy Dashboard

Data Elements:

- Medicine_Name, Medicine_ID
- Batch, Expiry_Date, Quantity
- Derived KPIs: Total Stock, Low Stock, Expiring Soon, Out of Stock
- Pending Orders

Design Choices:

- KPI summary cards with color coding
- Inventory table (Medicine, Batch, Expiry, Quantity)
- Filters (Category, Stock Level, Expiry)
- Sidebar navigation (Inventory, Orders, Alerts)
- Clear alignment (names left, numbers right)

UI Mockup 3

Customer Order Status

Order Status

1



XYZ Pharmacy



 **Order MRD-789012**
For Mr.XYZ

Confirmed

Order Items:

Medication A (x2)
Syringes (pack of 10) (x1)
Bandages (x3)

130 Taka

Track Order

View Details

2



ABC Pharmacy



 **Order MRD-789013**
For Mr.XYZ

Pending

Order Items:

Pain Reliever (100mg) (x1)
Gloves (box of 50) (x2)

250 Taka

Track Order

View Details

Core Output 3

Customer Order Status

Data Elements:

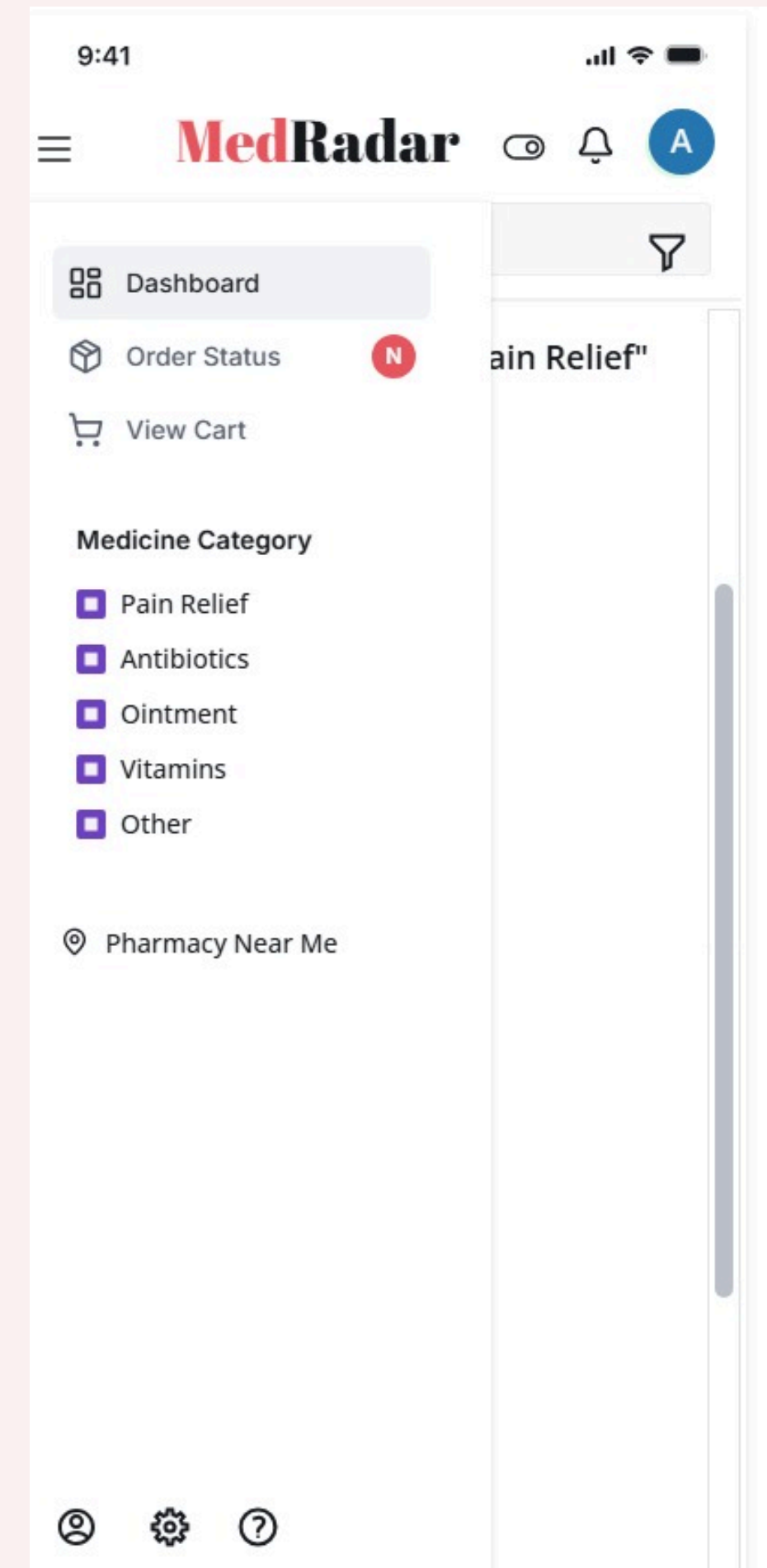
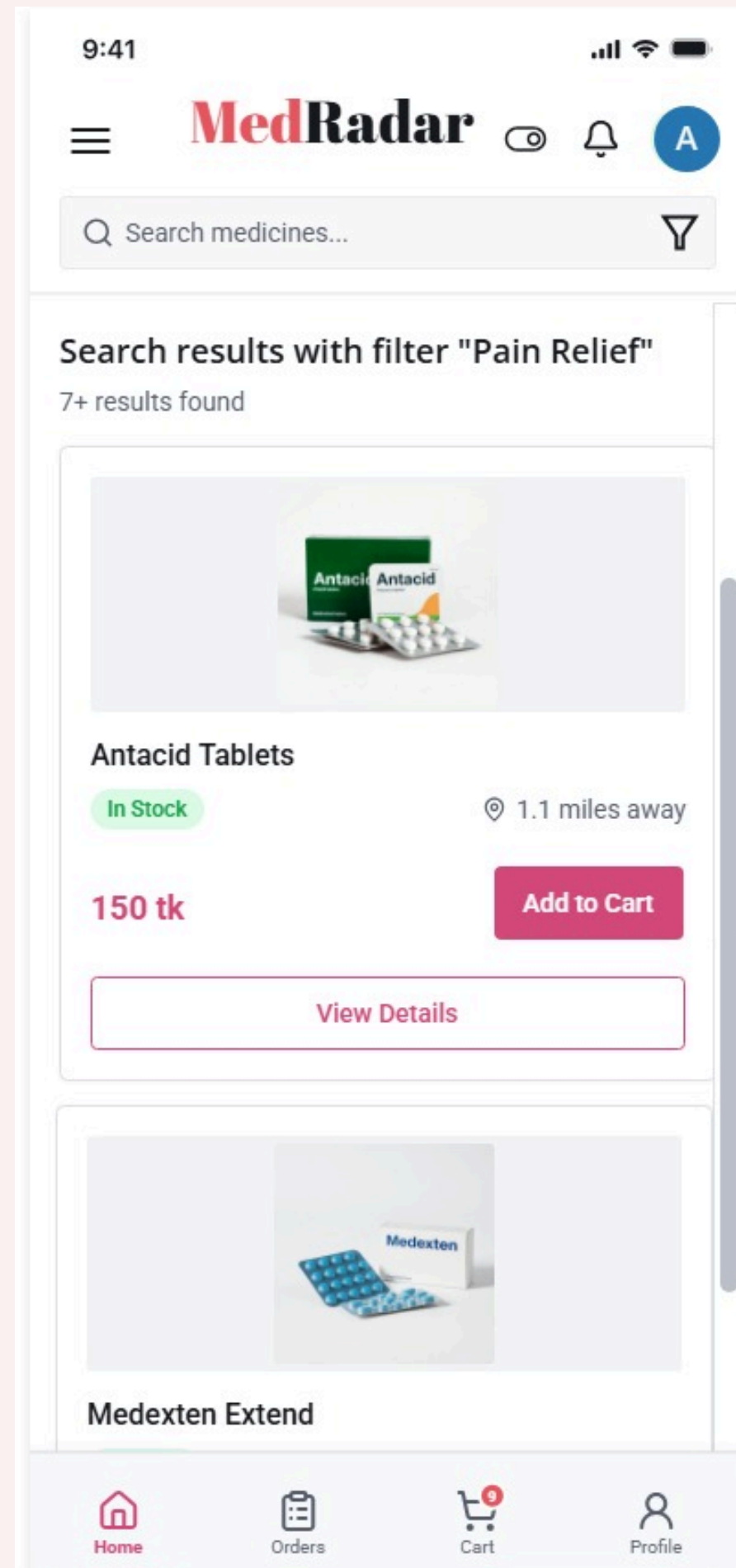
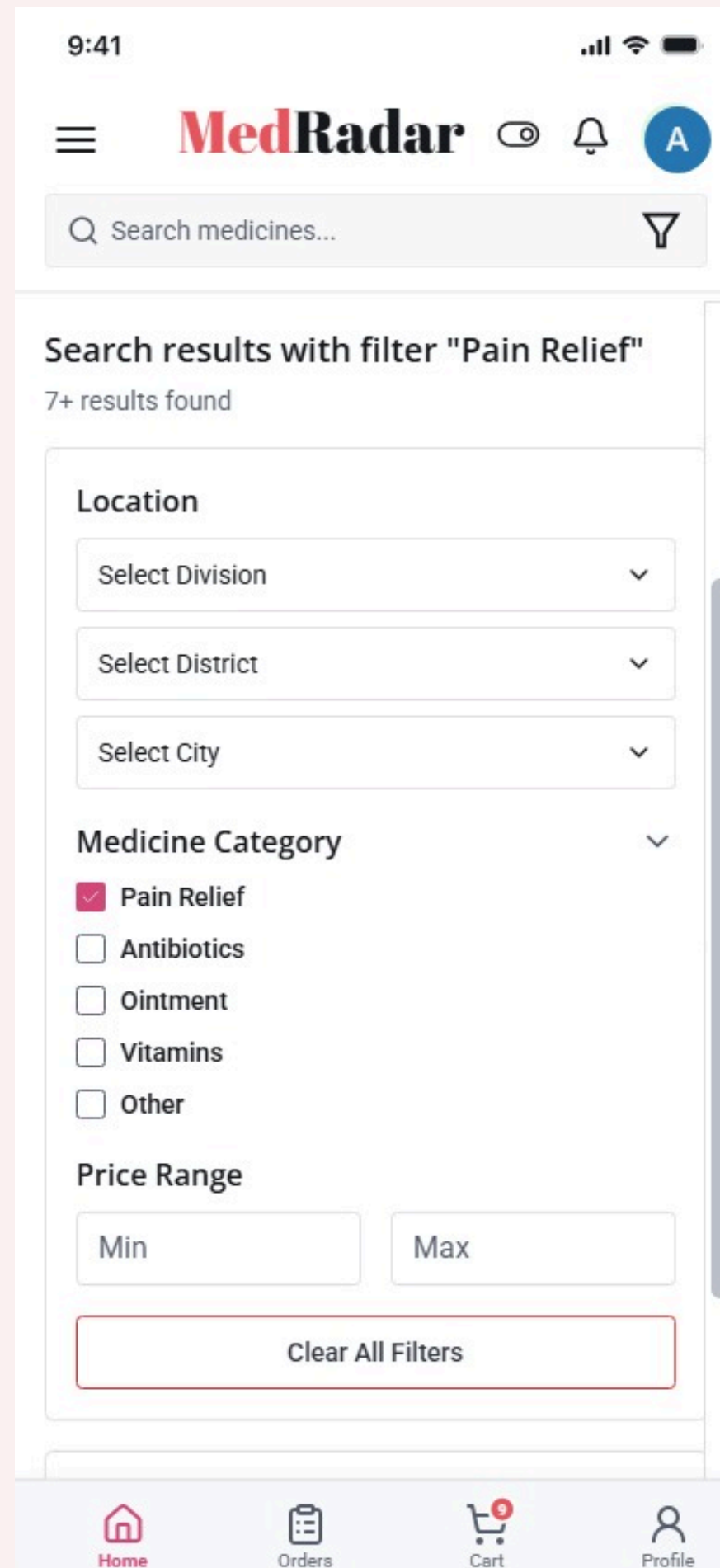
- Order_ID, Customer_ID
- Pharmacy_ID, Pharmacy_Name
- Order_Item_Record (Medicine_ID, Name, Qty)
- Order_Status (color-coded)
- Price/Total Amount

Design Choices:

- Order cards with pharmacy logo/name
- Status labels (Confirmed = green, Pending = orange, canceled- red)
- Clear item + price breakdown
- Action buttons (“Track Order”, “View Details”)
- Sidebar navigation (Orders, Cart, Notifications)

Mobile Interface UI

Customer Search Result



Mobile Interface UI

Medicine Search Result

Key Design Differences

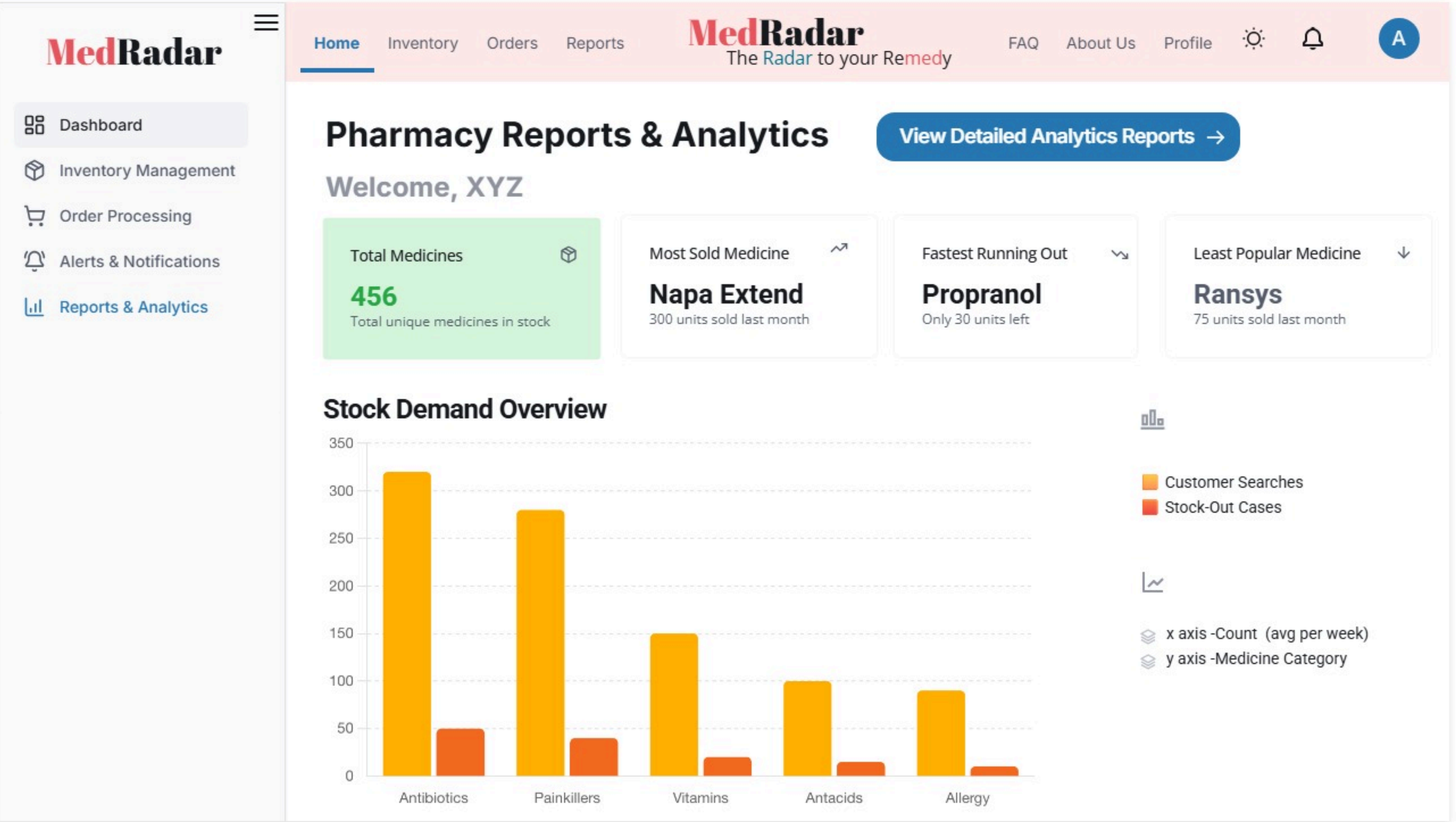
- Adapted to mobile layout.
- Collapsible Sidebar for better page view
- Footer for easy access to sections (Cart, Order, Home)
- More use of icons for easy and intuitive navigation

Data Visualization Output

Scenario:

Pharmacy users and managers often need to understand how medicine demand **changes over time** and which categories face the most shortages. Tabular transaction logs make it hard to see such trends and imbalances at a glance

Data Visualization Output



Data Visualization Output

Data Elements:

Medicine_Name, Category_Code, Quantity, Customer_Searches, Stock_Out Cases.

Design Choices:

1. KPI cards (Total Medicines, Most Sold, Fastest Running Out, Least Popular).
2. Bar chart comparing Customer Searches vs. Stock-Out Cases across categories.
3. Clear axis labels (x = category, y = count)
4. Color coding for easy distinction.

User Fit:

1. Helps pharmacies spot trends & shortages quickly.
2. Supports both restocking decisions and demand forecasting.

Justification for Design Choices

- **User-Centered** : Clean dashboards & filters for quick, intuitive tasks.
- **Highlighting** : Color coding (green, orange, red) for stock & order status.
- **Navigation** : Sidebars & filters group actions, reduce clicks.
- **Action-Oriented** : Clear CTAs (Add to Cart, Track, View) guide users.
- **Bias Control** : Sorting & filters (price, distance, stock) give fairness.
- **Visualization** : KPIs & charts reveal trends, shortages, performance.
- **Consistency** : Repeated layouts (cards, KPIs, filters) aid usability.

Thank you!

Feel free to ask any questions!